



phi4-reasoning

ollama run phi4-reasoning

Copy icon

↓ 631.2K ⌚ 6 months ago

Phi 4 reasoning and reasoning plus are 14-billion parameter open-weight reasoning models that rival much larger models on complex reasoning tasks.

14b

Models

[View all →](#)

9 models
phi4-reasoning:latest 11GB · 32K context window · Text · 6 months ago
phi4-reasoning:14b latest 11GB · 32K context window · Text · 6 months ago

Readme

Phi 4 reasoning and reasoning plus models are 14 billion parameter models that rival much larger models on complex reasoning tasks.

Phi 4 reasoning model is trained via supervised fine-tuning of Phi 4 on carefully curated reasoning demonstrations from OpenAI’s o3-mini. This model demonstrates meticulous data curation and high quality synthetic datasets allow smaller models to compete with larger counterparts.

Phi 4 reasoning plus model builds on top of Phi 4 reasoning, and is further trained with reinforcement learning to deliver higher accuracy.

Models

Phi 4 reasoning

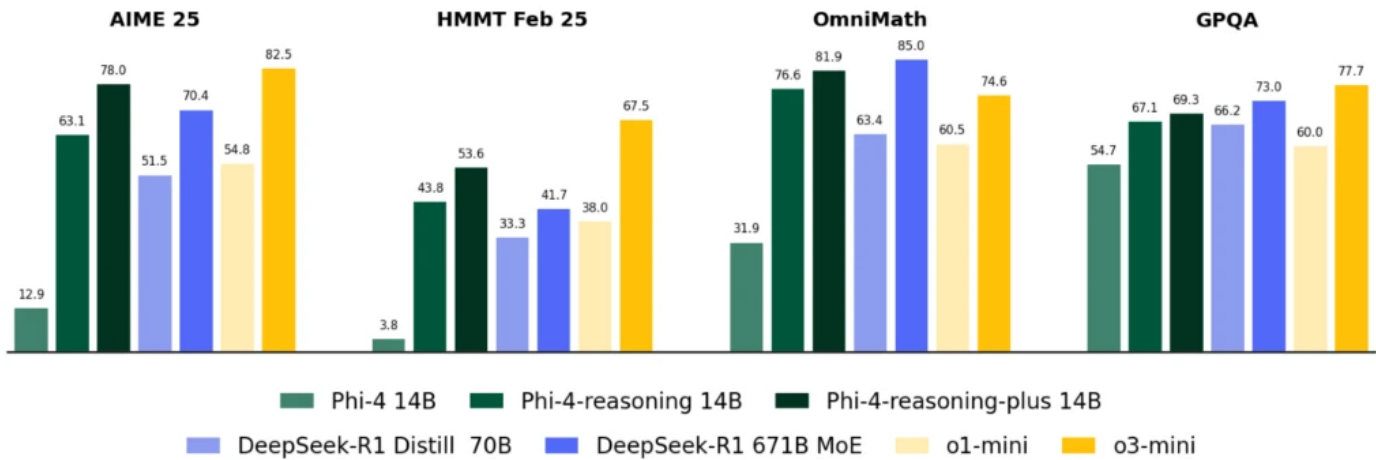
```
ollama run phi4-reasoning
```

Phi 4 reasoning plus

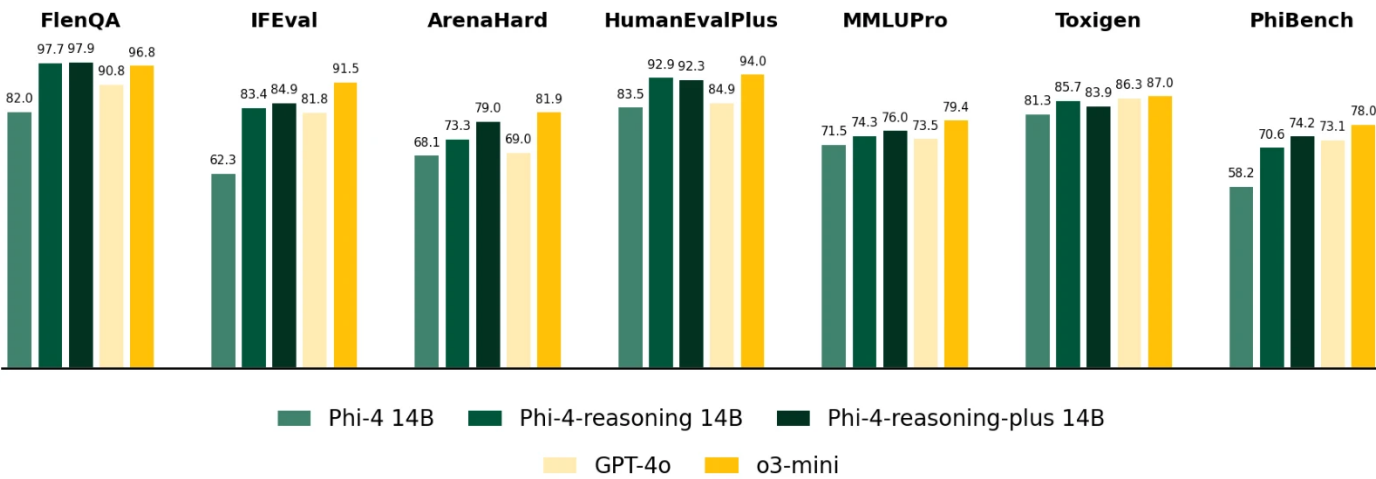
```
ollama run phi4-reasoning:plus
```

Benchmarks

Despite their small size (14B parameters), the performance of Phi-4-reasoning models exceeds or approaches models with 5-50x more parameters across several reasoning capabilities



Phi-4-reasoning performance across representative reasoning benchmarks spanning mathematical and scientific reasoning. We illustrate the performance gains from reasoning-focused post-training of Phi-4 via Phi-4-reasoning (SFT) and Phi-4-reasoning-plus (SFT+RL), alongside a representative set of baselines from two model families: open-weight models from DeepSeek including DeepSeek R1 (671B Mixture-of-Experts) and its distilled dense variant DeepSeek-R1 Distill Llama 70B, and OpenAI’s proprietary frontier models o1-mini and o3-mini. Phi-4-reasoning and Phi-4-reasoning-plus consistently outperform the base model Phi-4 by significant margins, exceed DeepSeek-R1 Distill Llama 70B (5x larger) and demonstrate competitive performance against significantly larger models such as Deepseek-R1.



Accuracy of models across general-purpose benchmarks for: long input context QA (FlenQA), instruction following (IFEval), Coding (HumanEvalPlus), knowledge & language understanding (MMLUPro), safety detection (ToxiGen), and other general skills (ArenaHard and PhiBench).

References

[Blog post](#)

[Blog](#) [Download](#) [Docs](#)

[GitHub](#) [Discord](#) [X \(Twitter\)](#) [Meetups](#)

© 2025 Ollama Inc.